

It is commonly used in South American herbal medicine for internal bleeding, particularly within the digestive tract—for example, rectal bleeding and hemorrhoids. It is also taken for bleeding in the urinary tract. Applied externally, a decoction of matico makes a valuable remedy for minor wounds, sore and inflamed skin, and insect bites and stings. The decoction may also be used as either a mouthwash or a douche.

Piper betle (Piperaceae) Betel

Description Slender climbing vine growing to 16 ft (5 m). Has heart-shaped leaves, tiny yellow-green flowers, and small spherical fruit.

Habitat & Cultivation Betel is native to Malaysia and southern India. It is widely cultivated in much of southern Asia, East Africa and Madagascar, and the Caribbean. The leaves are gathered throughout the year and dried for extracts or to use whole.



Betel leaves, traditionally chewed with areca nut and lime, give a mild sensation of well-being.

Parts Used Leaves, root, fruit.

Constituents Betel leaves contain up to 1% volatile oil (including cadinene, chavicol, chavibetol and cineole). As with many volatile oils, the percentages are variable. Malaysian samples have been shown to contain up to 69% chavibetol.

History & Folklore Betel leaves, wrapped around areca nut (*Areca catechu*) and lime (*Citrus aurantiifolia*), are known to have been chewed in India and Southeast Asia for several thousand years. Betel leaves are described in the *Mahavamsa*, the most ancient Sri Lankan text. Chewing quickly produces a red-stained saliva and does not, despite what is commonly thought, lead to blackened teeth. However, long-term use of betel leaves and areca nut is considered to increase

the incidence of cancer of the mouth and tongue. Ironically, the habit of chewing betel leaves is now giving way in many regions to cigarette smoking.

Medicinal Actions & Uses Betel leaves are chiefly used as a gentle stimulant, since they apparently induce a mild sensation of well-being. They also affect the digestive system, stimulating salivary secretions, relieving flatulence, and preventing worm infestation. In many Asian traditions, including Ayurvedic medicine, betel leaves are thought to have aphrodisiac and nerve tonic properties. In Chinese herbal medicine, betel root, leaves, and fruit are sometimes used as a mild tonic and stomach-settling herb. The root has been used with black pepper (*P. nigrum*, below) or jequirity (*Abrus precatorius*, p. 158) to produce sterility in women.

Caution The observed increase in the occurrence of oral cancers in regular users makes it unwise to chew betel.

Piper cubeba (Piperaceae) Cubeb

Description Climbing evergreen perennial growing to 20 ft (6 m). Has oval to oblong leaves, small flowers forming spikes, and round brown fruit.

Habitat & Cultivation Native to Indonesia, cubeb is cultivated in much of tropical Asia, especially in the shade of coffee bushes (*Coffea arabica*, p. 192). The fruit is gathered when immature.

Part Used Fruit.

Constituents Cubeb contains a volatile oil (up to 20%), a bitter principle (cubebin), an alkaloid (piperidine), resin, and fixed oil.

Medicinal Actions & Uses Like other members of the pepper family, cubeb has a significant antifatulent and antiseptic action. The fruit is used medicinally as a means to counter infections of the urinary tract, and has been taken in the past as a treatment for gonorrhea. In addition, the fruit is helpful in relieving digestive problems such as flatulence and bloating. Cubeb is occasionally employed as an expectorant in the treatment of chronic bronchitis.

Cautions Cubeb should not be taken by people suffering from kidney disease or inflammatory conditions of the digestive tract.

Piper nigrum (Piperaceae) Pepper

Description Perennial woody climber growing to about 16 ft (5 m). Has large oval leaves, spikes of small white flowers, and clusters of small round fruits, which ripen from green to red.

Habitat & Cultivation Native to southwestern India, pepper is now cultivated in tropical areas around the world. The fruit is harvested from plants that are at least 3 years old. Green peppercorns are picked unripe and pickled, black peppercorns are picked unripe and dried, red peppercorns are picked ripe and dried, and white peppercorns are picked ripe and soaked in water for 8 days before drying.



Pepper, here drying after the harvest, is still highly valued both medicinally and in cooking.

Parts Used Fruit, essential oil.

Constituents Pepper contains a volatile oil (including beta-bisabolene, camphene, beta-caryophyllene, and many other terpenes and sesquiterpenes), up to 9% alkaloids (especially piperine, which is largely responsible for the herb's acrid taste), about 11% proteins, and small quantities of minerals.

History & Folklore Cultivated as a spice and a medicine since ancient times, pepper was a vital commodity in world trade for thousands of years. Attila the Hun is reputed to have demanded 3,000 lb (1,360 kg) of pepper as ransom during his siege of the city of Rome (408 CE).

Medicinal Actions & Uses The familiar sharp taste of pepper reflects the stimulant and antiseptic effect it has on the digestive tract and the circulatory system. Pepper is commonly taken to warm the body, or to improve digestive function in cases of nausea, stomachache, flatulence, bloating, constipation, or lack of appetite. The essential oil eases rheumatic pain and toothache. It is antiseptic and antibacterial, and reduces fever.

Research Piperine, the main active constituent within black pepper, has significant therapeutic benefits, with a 2012 research paper listing "immunomodulatory, antioxidant, antiasthmatic, anticarcinogenic, anti-inflammatory, antiulcer and antiamebic properties." Piperine appears to aid the absorption of herbal and chemical medicines, e.g. curcumin (from turmeric, *Curcuma longa*, see p. 90), and in some cases, to slow their clearance by the liver.

Caution Do not take the essential oil internally without professional supervision.

Self-help Use Back pain, p. 313.